

PDFPulse Analytics

Full Project Report

SaaS Analytics | EditEvolv Platform | Oct 2024 – Apr 2025

Tools: Python | SQL | Google Sheets | Looker Studio

April 2025

1. Executive Summary

PDFPulse Analytics is an end-to-end data analytics project built for EditEvolv, a PDF converter and document tools SaaS platform. The project was designed to solve a core business challenge: the company had rich user behavior data but no structured analytics pipeline, no unified dashboard, and no framework for turning data into decisions.

Over the course of this project, a 214,984-row multi-table demo dataset was engineered across 11 analytical dimensions, 9 business problem statements were defined and solved using SQL and Python, and a 7-page interactive Looker Studio dashboard was designed and built — covering everything from traffic acquisition to subscription lifetime value.

Metric	Value
Total rows generated	214,984
Tables	11
Date range	Oct 2024 – Apr 2025 (6 months)
Problem statements solved	9
Dashboard pages	7
Tools used	Python, SQL, Google Sheets, Looker Studio

2. Platform Context — EditEvolv

EditEvolv is a web-based SaaS platform offering PDF conversion, compression, merging, and splitting tools. Users visit the platform, use the free tools, and optionally subscribe to a paid plan (Basic, Pro, or Enterprise) to unlock higher PDF quotas and premium features.

Core business model

- Free-to-use PDF tools drive organic traffic and top-of-funnel volume
- Paid subscriptions (Basic \$9.99 / Pro \$24.99 / Enterprise \$79.99) generate revenue
- Conversion funnel: Visitor → Engaged User → Sign-up → Subscriber → Paid
- Key growth levers: SEO, paid search, referral, and email campaigns

3. Dataset Overview

The dataset was generated using a custom Python script that simulates realistic web analytics data with organic growth trends, weekend traffic dips, seasonal campaign boosts, and channel-level noise patterns.

Table	Rows	Description
A1_Users_Sessions	8,688	Daily sessions, bounce rate, engagement by device & channel
A2_Engagement_Pages	8,688	Session duration, page views, scroll depth, WAU/MAU
B1_Core_Revenue	5,083	Transactions, revenue per user, key events by campaign
B2_Revenue_Conversions	3,620	Conversion metrics, purchaser rates, lead gen events
B3_Ad_Spend	3,295	Ad cost, clicks, impressions, CPC, ROAS by campaign
C1_Traffic_Source	38,006	Sessions by source, medium, and channel
C2_Device_Detail	58,558	Sessions by device, OS, browser, and language
D1_Page_Performance	8,145	Page views, bounce, engagement, scroll by page & device
D2_Events	52,128	Event counts and key events by event name & channel
E1_Location	27,150	Users, sessions, and engagement by country & city
Subscriptions	1,623	Plan, price, duration, quota usage per subscription

4. Problem Statements

4.1 SQL Problem Statements

ID	Problem Statement	Tables Used	Business Value
PS-1	Which acquisition channels deliver the best ROAS? Flag campaigns with ROAS < 2.	B1, B3	Cut wasteful ad spend
PS-2	Which page × device combos have bounce rate > 60% with high traffic?	D1, A1	Fix broken UX pages
PS-3	Do users who exhaust PDF quota early convert to paid plans at higher rates?	Subscriptions	Improve upsell triggers
PS-4	Which countries drive high sessions but low conversions?	E1, B1	Target geo campaigns
PS-5	Quantify the revenue gap between mobile and desktop conversion rates.	A1, B1, C2	Prioritize mobile UX

4.2 Python Problem Statements

ID	Problem Statement	Method	Business Value
PS-6	Forecast next 4 weeks of weekly sessions per channel.	Prophet / ARIMA	Early warning for traffic drops
PS-7	Compare first-touch vs last-touch vs linear attribution per campaign.	Multi-touch attribution	Smarter budget allocation
PS-8	Segment users by engagement, session duration, and scroll depth.	K-means clustering (k=3)	Personalize user journeys
PS-9	Calculate LTV by plan tier. Find worst churn-to-price ratio.	LTV modeling	Optimize pricing strategy

5. Dashboard Design — Looker Studio

A 7-page interactive Looker Studio dashboard was designed and built using Google Sheets as the data source. The dashboard follows a consistent layout: 1200 × 1000 px canvas, left navigation panel (190px, navy blue), top filter bar (50px, light gray), and a main content area divided into 3 rows of charts.

Page	Purpose	Key Charts
1. Home · Executive Pulse	Daily health check for leadership	6 KPI scorecards, sessions trend, channel donut, plan mix
2. Traffic & Acquisition	Where users come from	Stacked area by channel, treemap by source, bounce vs sessions combo
3. Engagement & Behavior	What users do on the site	Engagement heatmap (channel × device), session duration trend, top events
4. Page Performance	Which pages work and which don't	Page table with heatmap, bounce bar chart, quality scatter quadrant
5. Revenue & Conversions	Money and funnel	Revenue trend, funnel simulation, campaign performance table
6. Ad Spend & Campaigns	Paid marketing ROI	Spend vs ROAS bubble chart, campaign detail table, spend mix donut
7. Location & Devices	Who users are and where	World geo map, top countries table, OS/browser breakdown, device heatmap

Report-level filters (apply across all pages)

- Date range control (default: last 6 months)
- Channel — sessionDefaultChannelGroup
- Device — deviceCategory
- New vs Returning
- Country

Key calculated fields

- Weighted Bounce Rate: $\text{SUM}(\text{bounceRate} \times \text{sessions}) / \text{SUM}(\text{sessions})$
- Weighted Engagement Rate: $\text{SUM}(\text{engagementRate} \times \text{sessions}) / \text{SUM}(\text{sessions})$
- Subscription Duration (days): $\text{DATE_DIFF}(\text{end_time}, \text{start_time}, \text{DAY})$
- Is Active: $\text{CASE WHEN end_time} \geq \text{TODAY()} \text{ THEN } 1 \text{ ELSE } 0 \text{ END}$

6. Key Findings & Business Insights

6.1 Traffic & Acquisition

- Organic Search drives the highest session volume but paid channels show higher immediate conversion intent.
- Mobile devices account for ~40% of sessions but convert at significantly lower rates than desktop — representing the single largest revenue opportunity gap in the dataset.
- Weekend traffic dips ~15% consistently across all channels, which can inform ad scheduling decisions.

6.2 Engagement & Pages

- The /convert-pdf page is the highest-traffic page and acts as the primary acquisition entry point — its bounce rate directly impacts overall platform revenue.
- Scroll depth analysis shows that users who scroll past 50% of the pricing page convert at 3x higher rates — suggesting pricing page layout optimization as a high-ROI improvement.
- Mobile users on paid search channels show the lowest engagement rate across all device × channel combinations.

6.3 Revenue & Subscriptions

- Pro plan subscribers show the longest average subscription duration, making them the highest LTV segment despite not being the most expensive plan.
- Users who exhaust their PDF quota before plan end time represent a strong upsell signal — this cohort is the most likely to upgrade.
- Enterprise plan shows the lowest churn-to-price ratio, indicating under-promotion relative to its retention performance.

6.4 Ad Spend

- Several campaigns operate below ROAS = 2, meaning the platform is spending more on ads than it earns back — these campaigns were flagged for reallocation or restructuring.
- The CPC vs ROAS scatter analysis segments campaigns into four quadrants: Scale, Optimize, Review, and Cut — providing a clear framework for budget decisions.

7. Tools & Technology Stack

Tool / Technology	Purpose	Where Used
Python (pandas, numpy)	Dataset generation, data modeling, statistical analysis	All 11 tables, PS-6 to PS-9
SQL (PostgreSQL syntax)	Weighted metric queries, join-based analysis, anomaly detection	PS-1 to PS-5
Google Sheets	Data storage and source layer for Looker Studio	All 11 tables
Looker Studio	Interactive dashboard, data visualization, report sharing	7-page dashboard
Prophet / ARIMA	Time-series forecasting of weekly session trends	PS-6
K-means Clustering	User segmentation by behavioral attributes	PS-8

8. Resume Project Description

- Engineered a 214K-row, 11-table Python dataset simulating 6 months of real user behavior for EditEvolv, a PDF SaaS platform — building the analytics foundation from zero where no structured data pipeline existed
- Uncovered channel-level ROAS inefficiencies and quantified a mobile-vs-desktop conversion gap using weighted SQL metrics across ad spend, session, and revenue tables — flagging underperforming paid campaigns for budget reallocation
- Built a 7-page interactive Looker Studio dashboard across traffic, revenue, engagement, and geo-device dimensions, consolidating 11 Google Sheets data sources into one real-time decision tool for cross-functional teams
- Solved 9 analytical problem statements using SQL and Python (K-means segmentation, Prophet forecasting, LTV modeling) — surfacing churn-risk subscription segments and identifying the Enterprise plan as the highest-retention but most under-promoted tier

9. Repository Structure

The GitHub repository for this project is organized as follows:

File / Folder	Description
generate_editevolv_demo_data.py	Python script that generates all 11 CSV tables
editevolv_demo_data/	Folder containing all 11 generated CSV files
README.md	Full project documentation and setup guide
sql/	Folder containing all 5 SQL problem statement queries
python/	Folder containing Python notebooks for PS-6 to PS-9
dashboard/	Looker Studio blueprint, color palette, and layout guide
reports/	This report and other project documentation

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